

BTEC Level 3 Applied Science • Unit 1 Biology • Worksheet

Microscopy

Original JDSscience practice. These questions were written for this resource and are not official exam-board items.

Learning focus: magnification, resolution, conversions and instrument choice.

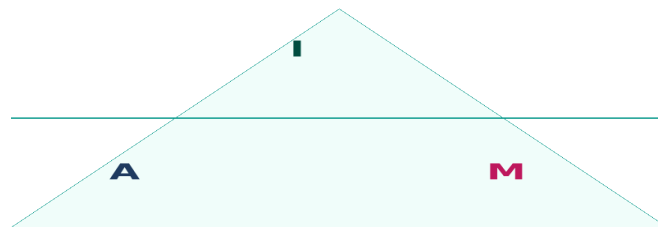
A Retrieval

1. Write the magnification formula and define resolution.

2. Convert 2.5 mm to μm and to nm.

Working

Magnification triangle



I = image size **A** = actual size **M** = I / A
Convert to the same unit first. 1 mm = 1000 μm = 1 000 000 nm

3. How do you calculate total magnification of a light microscope?

C Calculations — show every conversion

4. An image is 18 mm wide and magnification is $\times 600$. Calculate actual size in μm .

Working

5. A $7.5 \mu\text{m}$ cell is drawn 3.0 cm wide. Calculate magnification.

Working

6. A $10 \mu\text{m}$ scale bar measures 40 mm on a photograph. Calculate magnification.

Working

D Exam-style practice

7. Calculate the actual width of a cell if the image is 35 mm wide at $\times 500$. Give the answer in μm . [2]

Working

8. Compare the resolution of a light microscope and a TEM. [4]

9. Give two reasons for choosing a light microscope rather than an electron microscope. [2]

E Challenge

10. A student increases magnification from $\times 100$ to $\times 1000$ but cannot separate two close granules. Explain, using resolution.
